INCH-POUND

MIL-H-44428(GL)
7 August 1991

MILITARY SPECIFICATION

HOOD, GAS MASK, SPECIAL PURPOSE, TOXICOLOGICAL AGENTS

PROTECTIVE, M3

This specification is approved for use by the Natick Research, Development, and Engineering Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type and size of special purpose hood to provide military personnel protection against toxicological agents. The hood is to be used over a gas mask and Coverall, Toxicological Agents Protective, M3. This is a special purpose Life Support Clothing and Equipment (LSC&E) item. All Government administrative and surveillance procedures applicable to LSC&E items shall be invoked in accordance with the contract or purchase order (see 6.2).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be used in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5019 by using the Standardization Documents Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8415

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SPECIFICATIONS

FEDERAL

A-A-50083	-	Bag, Plastic, Folded Garment
V-T-285	-	Thread, Polyester
DDD-L-20	-	Label: For Clothing, Equipage and Tentage (General Use)
PPP-B-636	-	Boxes, Shipping, Fiberboard
PPP-T-45	-	Tape, Gummed, Paper, Reinforced and Plain, for Sealing and Securing

MILITARY

MIL-B-371	-	Braid, Textile, Tubular
MIL-C-484	-	Cloth, Wind Resistant Oxford, Cotton, Quarpel Treated
MIL-B-543	-	Buckles, Tongueless and Web Strap
MIL-F-10884	-	Fasteners, Snap
MIL-C-12189	-	Cloth, Coated: Butyl Coated, Toxicological Agents Protective
MIL-L-35078	-	Loads, Unit: Preparation of Semiperishable Subsistence Items; Clothing, Personal Equipment and Equipage, General Specification For
MIL-T-43566	-	Tape, Textile, Cotton, General Purpose, Natural or in Colors

STANDARDS

FEDERAL

FED-STD-191	-	Textile Test Methods
FED-STD-751	-	Stitches, Seams, and Stitching

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-147	-	Palletized Unit Loads
MIL-STD-731	-	Quality of Wood Members for Containers and Pallets

(Unless otherwise indicated, copies of federal and military specifications, standards and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the

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issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 2240 - Rubber Property - Durometer Hardness
D 3951 - Standard Practice for Commercial Packaging

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

(Non-Government standards and other publications are normally available from the organizations that prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

3.2 Guide samples. Samples, when furnished, are solely for guidance and information to the contractor (see 6.4). Variations from this specification may appear in the sample, in which case this specification shall govern.

3.3 Materials. It is encouraged that recycled material be used when practical as long as it meets the requirements of this specification.

3.3.1 Coated cloth. The coated cloth shall be butyl coated nylon cloth conforming to MIL-C-12189.

3.3.2 Bias binding. The bias binding for the snout rings shall be cut from the coated fabric specified in 3.3.1. As an alternate, the bias binding may be cut from oxford cloth conforming to type I, Olive Drab 7 of MIL-C-484.

3.3.3 Strapping. The strapping used to seal the seams shall be 3/4 +1/8, -0 inch wide and shall be made from the coated fabric specified in 3.3.1, straight or bias cut, except that a cotton-base cloth may be used in place of nylon provided the finished coated cloth meets the requirements of MIL-C-12189.

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3.3.4 Cement. The cement used to adhere the strapping to the seams shall be of the self-curing neoprene or butyl base type, pigmented to match the color of the coated cloth or clear. The cement, when applied and tested as specified in 4.4.1, shall not crack or flake when subjected to the cold crack test and the hydrostatic resistance of the cement coated cloth shall be not less than 90 percent of the actual hydrostatic value of the butyl coated cloth prior to cementing.

3.3.5 Thread, polyester. The thread for all stitching shall conform to type I, class 1, sub-class A of V-T-285, except that colorfastness requirements shall not apply. The size shall be as specified in table I. The thread shall be dyed Olive Drab S-1, C.A. 66022.

3.3.6 Tape, cotton. The cotton tape shall conform to type I, class 1a, 1/2 inch wide of MIL-T-43566, except the tape shall be dyed Olive Drab 7.

3.3.7 Braid. All braid shall conform to class 4 of MIL-B-371 and shall be dyed Olive Drab 7. The type shall be as specified below.

3.3.7.1 Braid for neck drawcord. The braid for the neck drawcord shall be type II. The ends of the braid shall be resin dipped or tipped 3/4 ± 1/8 inch to prevent raveling.

3.3.7.2 Braid for eye holes and snout drawcords. The braid for the eye holes and snout drawcords shall be type VII. The ends of the braid shall be stitched (see table I, operation 3c) or resin dipped or tipped 3/4 ± 1/8 inch to prevent raveling.

3.3.8 Buckle. The buckle shall be tongueless, one-bar, 5/8 inch, conforming to type I, style 2, class 1, construction B of MIL-B-543.

3.3.9 Snap fasteners. The snap fasteners shall conform to style 2A, finish 2 of MIL-F-10884. The size of the button and the eyelet components shall be selected by the contractor and shall be of sufficient length to provide a firm clinch.

3.3.10 Barrel lock. The barrel lock for the neck drawcord shall be rectangular, made of vinyl chloride polymer or copolymer and be pigmented Olive Drab or Black. It shall be sufficiently flexible for the purpose intended and shall have a hardness of 70 ± 10 when tested as specified in 4.4.1. The barrel lock shall measure 1-1/4 ± 1/16 inches by 3/4 ± 1/16 inch by 1/8 ± 1/16 inch. Two 1/4 ± 1/32 inch round holes shall be located on the center line parallel to the 3/4 inch dimension, with each hole 5/16 ± 1/16 inch from the end of the barrel lock. As an alternate, the barrel lock may be nexus cord lock number 194, color black, acetal polymer, or equal (see 6.5).

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3.3.11 Identification label. Each hood shall have an identification label conforming to type IV or VI, class 1 of DDD-L-20 with the following exceptions:

Type IV label (stamped)
(a) A vinyl base black indelible ink shall be used.
(b) Letters shall be sans-serif, extra bold, monotone style, 14 points (caps only).

Type VI label (cloth)
(a) The label shall be printed prior to impregnation or coating.
(b) Only chlorine bearing vinyl resin composition shall be used for impregnation or coating.
(c) The label shall measure 2-3/4 by 1-1/2 inches minimum and the legend shall be printed parallel to the small dimension.
(d) All colorfastness and writing tests are not applicable.

3.3.11.1 Resistance of labels to fading and tackiness. Both the cloth label and the stamped label shall be legible and shall not become tacky after testing as specified in 4.4.1.

3.4 Patterns. Standard patterns, which provide a seam allowance of 3/8 inch for all lapped seams unless otherwise specified in table I, will be furnished by the Government. The standard patterns shall not be altered in any manner and are to be used only as a guide for cutting the contractor's working patterns which shall be identical to the standard patterns.

3.4.1 List of pattern parts. The component parts for the hood shall be cut from the material specified and in the quantity indicated below:

Material	Item	Nomenclature	Cut parts
Cloth, coated, butyl coated, toxicological agents protective	1	Snout rings	2
	2	Front neck	1
	3	Back neck	1
	4	Front, left and right	2
	5	Back, left and right	2
	6	Snouts	2

3.5 Construction. The construction shall conform to the requirements specified in table I and herein. The hood shall be assembled with the heavily coated side of the cloth and strapping facing to the outside.

3.5.1 Stitches, seams, and stitching. All stitches, seams, and stitching shall conform to FED-STD-751. The type of seam, stitching and stitches per inch shall be as specified in table I. Seam allowances shall be maintained with seams sewn so that no raw edges, run-offs, twists, pleats, puckers, or open seams occur.

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3.5.1.1 Type 301 stitching. Unless otherwise specified, ends of all 301 stitching, except when caught in other seams or stitching, shall be backstitched or overstitched 3/8 inch minimum. Thread tension shall be maintained so that there will be no loose stitching resulting in loose bobbin or top thread, or excessively tight stitching resulting in puckering of the materials sewn. The lock shall be embedded in the materials sewn.

3.5.1.1.1 Repairs of type 301 stitching. Repairs of type 301 stitching shall be as follows:

a. When thread breaks or bobbin run-outs occur during sewing, the stitching shall be repaired by restarting the stitching a minimum of 3/8 inch back of the end of the stitching. 1/

b. Thread breaks or two or more consecutive runoff or skipped stitches noted during inspection of the item (in-process or end item) shall be repaired by overstitching. The stitching shall start a minimum of 3/8 inch back of the defective area, continue over the defective area and continue a minimum of 3/8 inch beyond the defective area onto the existing stitching. Loose or excessively tight stitching shall be repaired by removing the defective stitching, without damaging the materials, and restitching in the required manner. 1/

1/ When making the above repairs the ends of the stitching are not required to be backstitched.

3.5.1.2 Use of automated apparel equipment. Automated apparel equipment may be used to perform any of the operations specified in table I, providing that the seam and stitch type are as specified and the finished component conforms to the required configuration. When a government furnished shaper pattern is forwarded, the component shall conform to that pattern.

3.5.1.3 Thread ends. All thread ends shall be trimmed to a length of not more than 1/4 inch.

3.5.1.4 Bartacking. Bartacking shall be 1/2 ± 1/16 inch in length and 1/8 ± 1/32 inch in width and shall contain 28 stitches, unless otherwise specified. Bartacking shall be free from thread breaks or loose or tight stitching.

3.5.2 Application of seam strapping.

3.5.2.1 Preparation of seam area for strapping. Before applying the cement; the seam, the area along the seam where the strapping makes contact, and the surface of the strapping to be adhered, shall be thoroughly cleaned with rubber makers naptha or other suitable cleaning agent which is not a solvent for the butyl coating.

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3.5.2.2 Strapping of seams. The seams, where required, shall be strapped on both the inside and the outside using the cement specified in 3.3.4. Each individual seam shall be strapped with a continuous length of strapping with the strapping centered on the seam. Where ends of strapping meet, a 1 ± 1/4 inch overlap shall be made. Whenever strappings intersect, they shall be overlapped. When the strapping is centered over the seam and pressed into position, the cement line shall not extend more than 3/4 inch on either side of the strapping.

3.5.3 Setting of snap fasteners. A hole shall be prepunched to receive the button and eyelet components of the snap fasteners. The hole shall be smaller than the outside diameter of the button and eyelet barrels so that the barrel must be forced through the hole. The hole shall not be punched in the setting operation with the button or eyelet barrel. The fasteners shall be securely clinched without cutting the adjacent materials and no more than two splits shall occur in the button or eyelet barrel.

3.6 Manufacturing operations requirements. The hood shall be made in accordance with all operations listed in table I. The contractor is not required to follow the exact sequence of operations.

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NO.	TABLE I. MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
1.	Cutting. a. The hood components shall be cut in strict accordance with patterns furnished which show size and notches for the proper assembling of all parts. No pin tickets shall be used. The snout rings, straps, strapping, head loops and patches may be cut from ends. The use of drill holes is prohibited. b. The strapping shall be cut from the coated fabric and shall be 3/4 +1/8, -0 inch wide. c. The bias binding for the snout rings may be cut from either the coated fabric or cloth specified in 3.3.2. The width shall be 3/4 to 1-1/8 inches.						
2.	Replacement of defective components. During the spreading, cutting and manufacturing process, components having material defects or damages that are classified as defects in 4.4.3.1 and 4.4.3.2 shall be removed from production and replaced with non-defective and properly matched components.						
3.	Cutting fittings. a. Cut thirteen draw string loops from 1/2 inch cotton tape 1-1/2 ± 1/8 inches in length. b. Tubular braid, type II: Cut one length 50 ± 1/2 inches long for neck drawcord and either dip or tip both ends with resin to prevent raveling (see 3.3.7.1).						

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TABLE I. (cont'd)

NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
3.	Cutting fittings. (cont'd)						
	c. Tubular braid, type VII: Cut four pieces, two 20 ± 1/4 inches long for the snout drawcords and two 25 ± 1/4 inches long for the eyeholes. The ends of the braid shall be either finished with five rows of adjacent stitching or shall be dipped or tipped with resin to prevent raveling (see 3.3.7.2).	301	-----	8-10	B	B	
	d. Cut three head loops from basic material 2-1/2 ± 1/8 inches long and 1-1/8 ± 1/8 inches wide.						
	e. Cut 17 round patches from basic material 1-1/2 ± 1/8 inches in diameter.						
	f. Cut from basic material four 1/2 ± 1/16 inch by 1-3/4 ± 1/8 inches rectangular patches with corners rounded.						
4.	Make left and right snouts.						
	a. Turn under ends of four 1/2 inch wide tape loops 1/4 to 3/8 inch and bartack to each snout on outer surface in location shown on pattern ± 1/8 inch. The marks on pattern indicate the center of each loop. Orientation of the loops shall be such that the folded ends of each loop are parallel to the small end of the snouts after the seaming operation in 4.b. Bartack to be placed 1/8 +1/16, -0 inch from the turned under edges of the tape.	Bartack	-----	14 per bartack	50	50	
	b. With tape loops on inside surface superimpose side edges of snouts evenly on each other within 1/16 inch and stitch with row of stitching located 3/8 ± 1/16	301	LSq-2	8-10	B	B	

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					NEEDLE	BOBBIN/ LOOPER	COVER
4.	Make left and right snouts. (cont'd) inch from edge. Turn and top stitch with row of stitching located 1/16 to 1/8 inch from cut edges of snouts. or						
	c. As an alternate to the above, join side edges of snouts using seam type LSb-2 with 3/8 ± 1/16 inch seam allowance, stitch gauge 3/16 to 5/16 inch and stitch margin 1/16 to 1/8 inch from folded edge. Raw edges shall be on inside of snouts.	301	LSb-2	8-10	B	B	
	d. Bind the inner edge of the snout rings with the ends overlapped 1/4 inch minimum. Stitch 1/16 inch from inner edge of binding. If the cloth specified in 3.3.2 is used for the binding, the outer end shall be turned under 3/8 inch minimum.	301	BSc-1	8-10	B	B	
	e. With outer surface of snouts facing out superimpose outer edge of snout rings on the edge of the snouts at the smallest end. With the edges of the snouts and snout rings aligned within 1/16 inch stitch with row of stitching located 1/4 ± 1/16 inch from edge. Turn and top stitch 1/16 to 1/8 inch from folded edge of snout rings. NOTE: The seam formed above shall not be strapped.	301	LSq-2	8-10	B	B	

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					NEEDLE	BOBBIN/ LOOPER	COVER
5.	Form eyeholes in front pieces (typical left and right).						
	a. With inner surface of front piece facing up, position a 25 inch length tubular braid around the periphery of the eyehole. The braid shall be stitched around the periphery of the eyehole starting and stopping at the intersections of the eye tab. The free ends of the braid shall hang loose at the intersections of the eye tab 8 ± 1/2 inches. The edge of the braid shall extend beyond the edge of the eyehole 1/8 to 3/16 inch and the stitch margin shall be 1/16 to 1/8 inch from the edge of the eyehole.	301	SSa-1	8-10	B	B	
	b. Turn each side edge of eye tab back 1/8 to 3/16 inch and stitch 1/16 inch from the outer edge. NOTE: Operations 5a. and 5b. may be performed simultaneously with operation 5b. performed first.	301	EFa-1	8-10	B	B	
	c. Fold eye tab fully back onto inside surface of front piece and stitch across end 1/8 to 3/16 inch from edge of tab. d. Pass ends of braid through the tunnel formed by the eye tab.	301	EFa-1	8-10	B	B	
6.	Make three head loops. Apply cement across width of head loop strips and fold in half lengthwise with side edges aligned within 1/16 inch and press together firmly.						

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NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
7.	Assemble left and right fronts, snouts and back pieces.						
	a. With outside surfaces facing each other superimpose left and right fronts evenly, within 1/16 inch, along the edges for forming the center joining seam. Stitch along those edges with stitching located 1/4 to 5/16 inch from edges. During the stitching operation fold a head loop evenly in half within 1/16 inch and position between the fronts with the ends of the loop even with the front pieces and centered within 1/8 inch of the location shown on the pattern. Turn and top stitch with stitching located 1/8 ± 1/16 inch from folded edge.	301	LSq-2	8-10	E	E	
	b. With one snout turned inside out position around opening in front piece with notches aligned within 1/4 inch. Align edges of snouts and opening in front piece within 1/16 inch and stitch around periphery with stitch margin 1/4 ± 1/16 inch. Seam on snout should be placed below notch. NOTE: The outside surfaces of the snouts and front piece, where stitched together shall face each other.	301	LSq-2 (Step a)	8-10	E	E	
	c. Turn snouts right side out and top stitch 1/16 to 1/8 inch from folded edge of snouts.	301	LSq-2 (Step b)	8-10	E	E	
	d. With outside surfaces facing each other superimpose left and right back pieces evenly, within 1/16 inch, along the edges for forming the center joining	301	LSq-2	8-10	E	E	

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					NEEDLE	BOBBIN/ LOOPER	COVER
7.	Assemble left and right fronts, snouts and back pieces. (cont'd)						
	seam. Stitch back pieces together with stitching located 1/4 to 5/16 inch from edges. Turn and top stitch with stitching located 1/8 ± 1/16 inch from folded edge.						
	e. With the outside surfaces of the joined fronts and backs facing each other, superimpose the edges evenly within 1/16 inch and stitch along the edges with stitch margin 1/4 to 5/16 inch. During the stitching operation two head loops shall be caught in the seam. The loops shall be evenly folded in half within 1/16 inch and positioned between the front and back pieces with the ends even with the edges and centered within 1/8 inch of the locations shown on the pattern. Turn the joined front and back right side out and top stitch 1/8 ± 1/16 inch from folded edge.	301	LSq-2	8-10	E	E	
	f. Fold the bottom edge of the joined fronts and backs to the inside 1/4 to 3/8 inch and stitch 1/8 to 3/16 inch from the folded edge.	301	EFa-1	8-10	B	B	
8.	Make shoulder adjusting strap assemblies and attach to hood.						
	a. For short strap cut the basic material 4 ± 1/4 inches long by 1-1/8 ± 1/8 inches wide. Apply cement across width of strap, fold strap evenly in half across the width within 1/16 inch and press together firmly. Assemble strap to buckle and install female snap fastener.						

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					NEEDLE	BOBBIN/ LOOPER	COVER
8.	Make shoulder adjusting strap assemblies and attach to hood. (cont'd)						
	NOTE: As an option the folded strap may be stitched prior to installing fastener.						
	b. For long strap cut the basic material 19-1/2 ± 1/2 inches long by 1-1/8 ± 1/8 inches wide. Apply cement across width of long strap, fold strap evenly in half across the width within 1/16 inch, and press together firmly. Thread one end of strap around center bar of buckle and box stitch across width of strap with a box pattern 3/8 ± 1/16 inch long and 1/8 to 3/16 inch wide, or as alternate, bartack.	301 or Bartack	-----	8-10 28 per bartack	E E	E E	
	c. Assemble the short and long straps by threading the straps through the buckles.						
	d. Attach free ends of shoulder strap assemblies to back of hood in location shown on pattern, ± 1/8 inch with a 3/8 to 1/2 inch box - X stitch pattern. Location shown on pattern is the center of the width of the strap.	301	-----	8-10	E	E	
9.	Install male snap fasteners in hood.						
	Install two male snap fasteners to front of hood at bottom in location shown on pattern, ± 1/8 inch. Stud component shall be on outside surface.						

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					NEEDLE	BOBBIN/ LOOPER	COVER
10.	Strapping of seams and application of patches prior to attachment of neck component.						
	a. Securely cement and strap all inside and outside seams with strapping. Outside straps at sides, center, front and center back shall be turned over bottom hem and extend 1/4 inch minimum up onto inside surface of hood. Slit outside strapping to permit insertion of head loops. Hem at bottom of hood shall not be strapped.						
	b. On the inside surfaces of the hood components, cement the round patches over the back of the snap fasteners and attachment points for tape loops and adjusting straps. Cement the rectangular patches over the ends of the eyehole tabs and also the stitching on outside surface of hood.						
11.	Make neck component and assemble to hood.						
	a. With outer surfaces facing each other join the back and front neck pieces along the side edges. Edges shall be aligned within 1/16 inch and stitch margin shall be 1/4 to 5/16 inch. Turn and top stitch each seam with stitching located 1/8 ± 1/16 inch from folded edge.	301	LSq-2	8-10	E	E	
	b. Turn under ends of five 1/2 inch wide tape loops 1/4 to 3/8 inch and bartack to outer surface of neck component in location shown on pattern, ± 1/8 inch. The marks on pattern indicate the center of each loop. Bartack to be placed 1/8 +1/16, -0 inch from the turned under edges of the tape.	Bartack	-----	14 per bartack	E	E	

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					NEEDLE	BOBBIN/ LOOPER	COVER
11.	Make neck component and assemble to hood. (cont'd)						
	c. Securely cement and strap the side seams on the inside and outside surfaces.						
	d. Join neck component to inside of hood with row of stitching located 3/16 ± 1/16 inch from edge of neck. Top edge of neck shall be positioned in location shown on patterns within 1/8 inch with the back facing the back of the hood and the side seams of the hood neck components aligned within 3/8 inch.	301	SSa-1	8-10	E	E	
12.	Strapping of neck component attachment seam.						
	Securely cement and strap inside and outside of seam for attaching neck component to hood.						
13.	Insert snout drawcord.						
	Thread 20 inch length snout drawcords through the loops on the snouts with the free ends even and laying over the snouts joining seam. Knot ends together.						
14.	Insert neck drawcord.						
	Thread 50 inch neck drawcord through loops in neck-piece, draw both ends of cord side by side through each hole of the barrel lock, and knot ends together. Barrel lock to hang at front when threaded.						
15.	Attach label.						
	a. Stitch four edges of cloth label to a 3-1/4 by 2 inch piece of coated fabric.	301	SSa-1	8-10	E	E	

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NO.	TABLE I. (cont'd) MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
15.	Attach label. (cont'd) b. Cement label assembly to inside front of neck-piece near bottom. or c. The label may be stamped directly on the fabric in the same location.						
16.	Cleaning. Remove all stains and spots.						

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3.7 Performance.

3.7.1 Strapping adhesion. After application of the strapping as specified in 3.5.2, the strapping adhesion shall meet the requirements of table II when tested as specified in 4.4.1 and 4.4.4, as applicable.

TABLE II. Strapping adhesion

Characteristic	Minimum requirement
Adhesion, pounds per inch of width:	
Initial	3.0
After water immersion 1/	2.0
After heat aging 1/	2.0

1/ To be tested on sample seam. See footnote 2 under table III.

3.7.2 Leakage. The hood shall not leak when tested as specified in 4.4.4.

3.8 Functional performance. Conformance with design requirements as stipulated in this specification shall be determined on the basis of satisfactory functional performance of the hood, when the hood is worn over the M-9A1 gas mask. Functional performance is defined as follows:

a. The hood shall be compatible with the M-9A1 gas mask in all respects, i.e., all vision and respiratory component parts of the hood shall fit the counterpart components of the gas mask.

b. Accessory component parts of the hood, whose function is that of adjustment or closure, e.g., shoulder adjusting straps, snouts and eyehole closure components and studs and snap fasteners shall function readily so that the hood may be donned, adjusted, all closures made and the hood removed without difficulty or injury to the hood or gas mask.

3.9 Repairs in coating on finished hood. Areas containing coating defects may be repaired by application of two round, superimposed, 1-1/2 ± 1/4 inch diameter patches with one patch on the inside surface of the material and the other on the outside surface of the material. The patches shall be made of the coated cloth specified in 3.3.1 and shall be adhered with the cement specified in 3.3.4. The lightly coated side of the patch shall be cemented to the hood and the patch shall extend a minimum of 1/2 inch beyond all edges of the damaged area. The patch shall be smoothly and evenly adhered over its entire area, and shall be well adhered to the coated cloth so that, when subjected to flexing action between the hands, will show no lifting of the edge, and no cracking, flaking or removal of the cement or patch. The repaired area shall show no stiffening or tackiness which would effect the serviceability of the hood. Repairing of fabric coating defects by methods other than patching are not authorized.

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3.9.1 Location and number of repairs. Repairs not exceeding four in number may be made, provided not more than two repairs are made on any one of the six major pattern parts.

3.10 Workmanship. The end item shall conform to the quality of product established by this specification and the occurrence of defects shall not exceed the applicable acceptable quality levels. Utmost care shall be taken during component and subassembly fabrication to ensure quality workmanship and safety of the service person using the item.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Responsibility for dimensional requirements. Unless otherwise specified in the contract or purchase order, the contractor is responsible for ensuring that all specified dimensions have been met. When dimensions cannot be examined on the end item, inspection shall be made at any point or at all points in the manufacturing process necessary to ensure compliance with all dimensional requirements.

4.1.3 Certificates of compliance. When certificates of compliance are submitted, the Government reserves the right to inspect such items to determine the validity of the certification.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

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4.3 First article inspection. When a first article is required (see 3.1 and 6.2), it shall be examined for the defects specified in 4.4.3 through 4.4.3.3.1, and tested as specified in 4.4.4.

4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document. In addition, testing shall be performed on components and materials listed in table III for the characteristics shown. The inspection level for each component or material shall be S-1. The inspection level for characteristics to be tested on a composite basis shall be used only in determining the number of containers to be sampled in arriving at the composite sample. The lot shall be unacceptable if one or more units fail to meet any requirement specified. For those characteristics where test results are reported as pass or fail, there shall be no evidence of failure of any test specimen to meet the requirements specified. There shall be no failure to meet the lot average requirements as specified.

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TABLE III. Component tests

Component and lot expressed in terms of	Characteristic	Specification reference		Requirements applicable to		Number determinations per unit	Results reported as		Sample unit
		Requirement	Test method	Individual unit	Lot aver		Pass or fail	Numerically to nearest	
Cement (1 gallon)	Material identification	3.3.4	1/	-	-	-	-	-	-
	Cold crack at -20°F	3.3.4	4.5.2	X	-	1	X	-	1 pint composite and 1 yard of coated cloth
	Hydrostatic resistance	3.3.4	4.5.3	-	X	5	-	1 percent psi	
	Adhesion after water immersion	3.7.1	4.5.1.2 2/	-	X	5	-	0.1 lb. per inch of width	
	Adhesion after heat aging	3.7.1	4.5.1.3 2/	-	X	5	-	0.1 lb. per inch of width	
Barrel lock (1 each)	Material identification	3.3.10	1/	-	-	-	-	-	-
	Hardness	3.3.10	ASTM D 2240	X	-	2	-	1 unit	1 barrel lock
Label, cloth (when applicable- 1 gross)	Identification of impregnating material	3.3.11	1/	-	-	-	-	-	-
	Resistance to fading and tackiness	3.3.11.1	4.5.5	X	-	2	X	-	2 labels
Ink (applicable when marking is applied directly to end item - 1 gallon)	Resistance to fading and tackiness	3.3.11.1	4.5.5	X	-	2	X	-	2 labels 3/

1/ A certificate of compliance shall be submitted and will be acceptable for the stated requirement.

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2/ Two 4 inch wide by 1 yard long samples of coated cloth specified in 3.3.1 shall be joined using seam and stitch type specified in operation 7.a. of table I. The seamed samples shall then be strapped on one side only with a 1 inch strip of strapping material specified in 3.3.3 utilizing the cement specified in 3.3.4 and using the same procedure used during the strapping of the end item.

3/ A piece of the coated cloth with the legend stamped thereon shall be considered as one label. The coated cloth need only be large enough to include the legend.

4.4.2 In-process inspection. Inspection of the hood and its components shall be made during the manufacturing operations to ascertain that construction details which can not be examined on the finished product are in accordance with specified requirements. The in-process inspection shall include inspection of the working patterns to assure that they conform to the standard patterns, inspection of the cut parts covered by patterns to ascertain that the parts are free of material defects and damages that are classified as defects in 4.4.3.1 and are cut properly with respect to size, that stitching for attaching neck components is correct and that the diameter of the holes for the snap fasteners is smaller than the outside of the button and eyelet barrels. Whenever nonconformance is noted, corrections shall be made to the components affected and lot in-process. Parts which can not be corrected shall be removed from production.

4.4.2.1 Inspection of hood components prior to strapping. The neck component prior to strapping of seams and prior to attachment to the hood with attached shoulder adjusting strap assemblies, and the hood prior to strapping of seams shall be examined for the defects listed in table IV. The lot size shall be expressed in units of individual hoods with attached shoulder adjusting strap assemblies with unattached neck component. The sample unit shall be one hood with attached shoulder adjusting strap assemblies with unattached neck component. The inspection level shall be S-4 and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 4.0 for major defects and 15.0 for total (major and minor combined) defects.

TABLE IV. Hood component defects prior to strapping

Examine	Defect	Classification Major	Minor
Open seam	1/2 inch or less		201
	More than 1/2 inch	101	
	NOTE: A seam shall be classified as open when one or more stitches joining a seam are broken or when two or more consecutive runoffs or skipped stitches occur.		

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TABLE IV. Hood component defects prior to strapping (cont'd)

Examine		Defect	Classification Major	Minor
Runoff (see open seam)				
Seam and stitch type		Wrong seam or stitch type	102	
Stitch tension		Loose, resulting in loose bobbin or top thread		202
		Excessively tight, resulting in puckering of material		203
Stitches per inch		One to two stitches less than minimum specified		204
		Three or more stitches less than minimum specified	103	
		More than the maximum number of stitches per inch:		
		- but not weakening material		205
		- resulting in weakening of material	104	
Thread breaks, skipped stitches or runoff (on type 301 stitching)		Not overstitched as specified		206
	NOTE:	Thread breaks or two or more consecutive runoffs or skipped stitches not overstitched shall be classified as open seams.		
Stitching ends (on type 301 stitching)		Ends of stitching not secured as specified (except when caught in other stitching or turned under in a hem)		207
Rows of stitching		Any row missing	105	
Needle chews		Any needle chew	106	
Stitch margin		Not within specified tolerance		208
Bartacks		Any bartack omitted	107	
		Any bartack not as specified or not in specified location		209
		Loose stitching, incomplete or broken stitching		210

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4.4.2.2 End item critical defect examination. Every end item, prior to the examinations in 4.4.3, shall be examined for the critical defects listed in table V. Any end item found to contain one or more critical defects shall be rejected.

4.4.3 End item examination.

4.4.3.1 End item visual examination. The end items shall be examined for the defects listed in table V. The lot size shall be expressed in units of hoods. The sample unit shall be one hood. The inspection level shall be II and the AQL, expressed in terms of defects per hundred units, shall be 2.5 for major defects and 15.0 for total (major and minor combined) defects. The finding of one or more critical defects shall be cause for rejection of the lot.

TABLE V. End item visual defects

Examine	Defect	Classification Critical	Major	Minor
Braid or tape	Cut edge		101	
	Frayed edge			201
Coated fabric, including strapping and patches	Cut, tear, or hole	1		
	Abrasion mark or scratch which extends through the coating to a point where base fabric yarns or fabric weave can be seen	2		
	Uncoated area	3		
	Pit, blister, tunnel, or delamination of coating	4		
	Lump or imbedded foreign matter which protrudes so as to be readily abraded from material or easily removed	5		
	Crease or wrinkle which when flattened by pulling manually results in delamination of coating from base fabric	6		
	Spot, stain, or streak more than 1 inch in combined direction which cannot be readily removed with naptha or water			202
	Brittle area which when bent results in a crack	7		

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TABLE V. End item visual defects (cont'd)

Examine		Defect	Classification Critical	Major	Minor
Coated fabric, including strapping and patches (cont'd)	NOTE:	A pick glass may be used to determine the degree of deficiency. If any of the above conditions occur on a reinforcement patch, hem or adjustment strap, it shall be classified as a minor defect.			
Hardware (general)		Broken or malformed, corroded area, burr or sharp edge		102	
		Finish omitted			203
Snap fasteners		Any fastener not functioning properly, i.e., fails to snap closed, provide a secure closure or to open freely		103	
		Clinched excessively tight, cutting adjacent material		104	
		Clinched loosely, permitting either component to rotate freely:			
		- but not to the degree that either component can be expected to become detached during use			204
		- to the degree that components can be expected to become detached during use		105	
		More than two splits in eyelet or button barrel			205
		Incorrect style		106	
	NOTE:	The fasteners shall be snapped and unsnapped twice to determine whether parts of fastener separate freely and also effect a secure closure.			
Buckle		Not as specified		107	
Design and construction		Any characteristic not in accordance with specified requirements	8		

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TABLE V. End item visual defects (cont'd)

Examine	Defect	Classification Critical	Major	Minor
Design and construction (cont'd)	Heavily coated side not on exterior of hood		108	
	Any reinforcement patch omitted		109	
Repairs	Patch diameter less than 1-1/4 inches		110	
	Patch diameter greater than 1-3/4 inches			206
	More than 2 patches per panel			207
	More than 4 patches per hood			208
	Damaged area patched on one side only		111	
	Heavily coated side of patch cemented to the hood			209
	Patch extends less than 1/2 inch beyond all edges of the damaged area		112	
	Patches not superimposed by more than 1/4 inch			210
	Edges of patch separate from hood or cement cracks or flakes when hand flexed as specified in 3.9 resulting in edges of patch not adhered:			
	- for a distance of 1/16 inch inward and 1/2 inch or more around periphery of patch			211
	- for a distance of more than 1/16 inch but not more than 3/16 inch inward and 1/2 inch or more around periphery of patch		113	
	- for a distance of more than 3/16 inch inward regardless of distance around periphery of patch	9		
	Patch not smoothly and evenly adhered		114	
	Patch area excessively stiff or tacky affecting serviceability	10		

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TABLE V. End item visual defects (cont'd)

Examine	Defect	Classification Critical	Major	Minor
Repairs (cont'd)	Any unauthorized repair, e.g., coating or cement applied to defect in fabric coating in lieu of patching	11		
Component and assembly	Any component omitted, damaged, not as specified, not firmly affixed or misplaced to such a degree that:			
	- injury to wearer may result	12		
	- serviceability is seriously affected		115	
	- serviceability is affected but not seriously			212
	Any end of braid not resin dipped, stitched, or tipped as applicable			213
Strapped seams	Any open seam resulting in a through hole	13		
	Any seam strapped on only one side	14		
	Any loose strapping, i.e., any area not cemented unless otherwise specified herein	15		
	Any blister or delamination in central area of strapping but edges are securely affixed and the bonded portion of strapping is securely cemented to fabric			214
	Strapping on one side does not completely cover stitching on seam		116	
	Strapping not centered			215
	Width of strapping not uniform			216
	Piecing of strapping on individual seams		117	

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TABLE V. End item visual defects (cont'd)

Examine	Defect	Classification Critical	Major	Minor
Strapped seams (cont'd)	Any ends of strapping that meet, not overlapped 1 ± 1/4 inch		118	
	Seam area covered by strapping not smooth and free from puckers or twists		119	
	Any opening at edge of strapping 1/16 inch or more but does not extend to stitching			217
	Any opening extending to the stitching:			
	- on one side only		120	
	- on both sides in same area	16		
	NOTE: Extent of opening (or tunnel) may be determined by the insertion of any blunt end instrument suitable for this purpose. Penetration of the instrument shall cease immediately if resistance is met.			
Unstrapped seams	Any open seam, skipped stitches or run-off stitches on hem:			
	- 1 inch or less			218
	- more than 1 inch		121	
	Any open seam, skipped stitches or run-off on the binding seam or snout ring seam:			
	- 1/2 inch or less			219
	- more than 1/2 inch		122	
Workmanship	Cement line extends more than 3/4 inch on either side of strapping on major portion of seam			220
	Any cemented area not fully cured		123	
Marking or labels	Missing, incomplete, incorrect, not legible, not specified type or size, not in proper location, or not accomplished in the specified manner			221

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4.4.3.2 End item dimensional examination. The end item shall be examined for conformance to the dimensions specified in table I, including stitch margins, that can be examined on the end item. Any measurement deviating from the specified dimension shall be classified as a defect. The lot size shall be expressed in units of hoods. The sample unit shall be one hood. The inspection level shall be S-3 and the AQL, expressed in terms of defects per hundred units, shall be 4.0.

4.4.3.3 Examination of hood for functional performance. The hood shall be examined to determine conformance with design requirements. The lot size shall be expressed in units of hoods. The sample unit for this examination shall be one completely fabricated hood. The inspection level shall be S-3 and the AQL, expressed in terms of defects per hundred units, shall be 1.5 for major defects and 6.5 for minor defects. The procedure to be employed in conducting this examination shall be as follows:

a. The hood shall be placed over the gas mask and necessary adjustments and closure made. This involves the following operations: Secure the eyeholes around the goggles portion of the mask by means of the tubular braid, which should be drawn taut, knotted and then allowed to fall freely inside the face portion of the hood; secure the snouts to the gas mask cannister by means of the snout drawcords.

b. During this process, the inspector shall note the following details:

(1) Rupture, deformation, loosening or detachment of any component during adjustments or closure operations.

(2) Misalignment of eyeholes causing impairment of vision through the mask goggles.

(3) Obstruction of the cannister caused by misfit of the snouts on the cannister.

(4) Slippage of the cannister through the snout rings.

c. The hood shall be removed from the gas mask by the reverse operations detailed in "a." above.

d. During removal of the hood, the inspector shall note details in reverse order as outlined in b. (1) and b. (2).

4.4.3.3.1 Classification of defects. Defects in functional performance shall be classified as Major or Minor as follows: Major, encompasses defects as described in 4.4.3.3 in relation to functional performance defined in 3.8.a. Minor, encompasses defects as described in 4.4.3.3 in relation to functional performance defined in 3.8.b.

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4.4.4 End item testing. The end items shall be tested for the characteristics listed in table VI. The lot size shall be expressed in units of hoods. The sample unit shall be two completely fabricated hoods. The inspection level shall be S-2. The lot shall be unacceptable if one or more sample units fail to meet any requirement specified.

TABLE VI. End item tests

Characteristic	Requirement paragraph	Test method	Number of determinations per sample unit	Results reported as
Initial adhesion of strapping	3.7.1	4.5.1.1	5	Numerically to nearest 0.1 lb.
Leakage	3.7.2	4.5.4	1	Pass or fail

4.4.5 Packaging examination. The fully packaged end items shall be examined for the defects listed below. The lot size shall be expressed in units of shipping containers. The sample unit shall be one shipping container fully packaged. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

Examine	Defect
Marking (exterior and interior)	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application
Materials	Any component missing, damaged, or not as specified
Workmanship	Inadequate application of components, such as: incomplete sealing or closure of flap, loose strapping, inadequate stapling, or improper taping Bulged or damaged container
Content	Number of hoods per intermediate pack is more or less than specified 1/ Number of hoods per shipping container is more or less than specified

1/ For this defect, one intermediate pack in the sample shall be examined.

4.4.6 Palletization examination. The fully packaged and palletized end items shall be examined for the defects listed below. The lot size shall be expressed in units of palletized unit loads. The sample unit shall be one palletized unit load, fully packaged. The inspection level shall be S-1 and the AQL, expressed in terms of defects per hundred units, shall be 6.5.

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Examine	Defect
Finished dimensions	Length, width, or height exceeds specified maximum requirement
Palletization	Pallet pattern not as specified Load not bonded with required straps as specified
Weight	Exceeds maximum load limits
Marking	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application

4.5 Methods of inspection.

4.5.1 Adhesion of strapping.

4.5.1.1 Initial adhesion. After the specimen has aged 48 hours, the adhesion of strapping shall be determined as specified in Method 5962 of FED-STD-191, except that adhesion shall be the average of the three lowest individual load resistance registered in pounds on the autographic chart divided by the width of the seam in inches. Each of the five test specimens should be taken from a different part of the item. The adhesion of strapping should be conducted on the heavily coated side (outside) of the garment.

4.5.1.2 Adhesion after water immersion. The specimen shall be immersed for 2 hours in boiling water, removed, immersed for 15 minutes in water at $75^{\circ} \pm 5^{\circ}F$, removed from the water and while still dripping wet, tested for strapping adhesion as specified in 4.5.1.1.

4.5.1.3 Adhesion after heat aging. The specimen shall be exposed for 7 days in a circulating air oven at $158^{\circ} \pm 2^{\circ}F$, removed from the oven, conditioned for 24 hours at $70^{\circ}F \pm 2^{\circ}F$ and 64 $\pm$ 2 percent relative humidity and then tested for strapping adhesion as specified in 4.5.1.1.

4.5.1.4 Test conditions. Results of physical tests obtained under testing conditions defined in FED-STD-191 and ASTM D 2240 will be acceptable except in dispute cases. In dispute cases, tests shall be conducted with both the specimen and test apparatus under standard conditions as defined in FED-STD-191.

4.5.2 Cold crack of cement film. Test specimens of the coated cloth specified in 3.3.1, measuring 8 by 8 inches minimum, shall be cleaned on one side only by wiping with rubber makers naptha or other suitable cleaning agent which is not a solvent for butyl coating. The cleaned area shall then be coated with two layers of the cement specified in 3.3.4. Test specimens shall then be conditioned for 4 hours at room temperature and dusted until tack free.

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4.5.2.1 Procedure. Specimens shall be tested as specified in Method 5874 of FED-STD-191 except that testing temperature shall be -20°F and specimens shall be exposed to this temperature for not less than 4 hours before testing. When performing tests, the cement side shall be considered the heavily coated side.

4.5.3 Hydrostatic resistance. Ten 8 inch by 8 inch specimens shall be randomly selected from a 1 yard piece of the coated cloth specified in 3.3.1. Five specimens shall be tested for hydrostatic resistance as specified in Method 5512 of FED-STD-191. The remaining five specimens shall be cement coated as specified in 4.5.2 and tested for hydrostatic resistance in accordance with Method 5512. The side without the cement coating shall be exposed to the water.

4.5.4 Leakage test. Press the top of the hood down to form a pocket approximately 10 inches in diameter and 6 inches deep. The eyepiece openings shall form part of the rim of the pocket. Pour water carefully into this pocket without getting any of the water into the eyepiece opening. Let the water stand for 5 minutes. Carefully empty the pocket without spilling any water on the inside of the hood. Turn hood inside out with dry hands and inspect for evidence of water leakage.

4.5.5 Resistance of label to fading and tackiness.

4.5.5.1 Boiling water immersion. Completely submerge one label in boiling water for a period of 1 hour. Remove and examine for legibility and tackiness.

4.5.5.2 Acetic acid immersion. Completely submerge one label in 10 percent (by volume) acetic acid solution at 75° to 80°F for 1 hour. Remove and examine for legibility and tackiness.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or Commercial, as specified (see 6.2).

5.1.1 Level A preservation. Each hood shall be laid face up with bottom edges aligned and the straps folded over the hood. The hood shall then be inserted in a snug-fitting flat or square clear polyethylene film bag conforming to A-A-50083. Fifteen hoods, alternately reversed, shall be placed flat in an intermediate fiberboard box conforming to style RSC, type CF, class domestic, variety SW, grade 200 of PPP-B-636. Inside dimensions of each fiberboard box shall be 19-1/2 inches in length, 29-1/2 inches in width, and 10 inches in depth. Box closure shall be secured with 2-inch minimum width gummed paper tape conforming to type III, grade B of PPP-T-45.

5.1.2 Commercial preservation. Hoods shall be preserved in accordance with ASTM D 3951.

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5.2 Packing. Packing shall be level A, B or Commercial, as specified (see 6.2).

5.2.1 Level A packing. Hoods, preserved as specified in 5.1, shall be packed in the quantity and arrangement as specified in table VII within a fiberboard shipping container conforming to style RSC-L, grade V2s of PPP-B-636. Approximate inside dimensions shall be as specified in table VII. Each shipping container shall be closed in accordance with method III, waterproofed in accordance with method V and reinforced as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.4.5. Shipping containers shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified (see 6.2). Strapping shall be limited to nonmetallic strapping, except for type II, class F loads.

5.2.2 Level B packing. Hoods, preserved as specified in 5.1, shall be packed in the quantity and arrangement as specified in table VII within a fiberboard shipping container conforming to style RSC-L, type CF (variety SW) or SF, class domestic, grade 275 of PPP-B-636. Approximate inside dimensions shall be as specified in table VII. Each shipping container shall be closed in accordance with method II as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.4.5.

5.2.2.1 Weather-resistant fiberboard containers. When specified (see 6.2), the shipping container shall be a grade V3c, V3s, or V4s fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with method III as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.4.5.

TABLE VII. Hood quantity and arrangement within each container and dimensions

Quantity per container (each)	Arrangement of level A intermediate boxes	Approximate inside dimensions of container (inches) 1/
30	Flat, two in depth	20-1/4 x 20-1/4 x 21-1/2

1/ Approximate dimensions offered as a guide only.

5.2.3 Commercial packing. Hoods, preserved as specified in 5.1, shall be packed in accordance with ASTM D 3951.

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5.3 Palletization. When specified (see 6.2), hoods, packed as specified in 5.2.2 or 5.2.3 shall be palletized on a 4-way entry pallet in accordance with load type Ia of MIL-STD-147. Pallet type shall be type I (4-way entry), type IV, or type V in accordance with MIL-STD-147. Pallets shall be fabricated from wood groups I, II, III, or IV of MIL-STD-731. Each prepared load shall be bonded with primary and secondary straps in accordance with bonding means C and D or film bonding means F or G. Pallet pattern shall be number 6 in accordance with the appendix of MIL-STD-147.

5.4 Marking. In addition to any special marking required by the contract or purchase order, unit packs, shipping containers, and palletized unit loads shall be marked in accordance with MIL-STD-129 or ASTM D 3951, as applicable.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The hood is intended for use over the M-9A1 Mask and Coverall, Toxicological Agents Protective, M3. The ensemble provides protection against toxicological agents.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Special provisions for verification inspection of Life Support Clothing and Equipment (see 1.1).
c. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
d. When a first article sample is required (see 3.1, 4.3 and 6.3).
e. Levels of preservation and packing (see 5.1 and 5.2).
f. Type and class of unit load specified (see 5.2.1).
g. When weather-resistant grade fiberboard shipping containers are required for level B packing (see 5.2.2.1).
h. When palletization is required (see 5.3).

6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209-4. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should also include specific instructions in all acquisition instruments regarding arrangements for selection, inspection, and approval of the first article.

6.4 Samples. For access to samples address the contracting activity issuing the invitation for bids or request for proposal or as directed by the contracting activity.

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6.5 Equal item. Prior to the use of an "or equal item", the supplier shall submit the item with supporting data to the contracting officer for subsequent approval or disapproval by the responsible military agency.

6.6 Subject term (key word) listing.

Chemical protection
Headgear
Life Support
LSC&E
M-9A1 gas mask

Custodian:

Army - GL

Preparing activity:

Army - GL

(Project 8415-A844)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-H-44428(GL)	1991 August 7

3. DOCUMENT TITLE
HOOD, GAS MASK, SPECIAL PURPOSE, TOXICOLOGICAL AGENTS PROTECTIVE, M3

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)	
U.S. Army Natick RD&E Center	(1) Commercial 508-651-4532	(2) AUTOVON/DSN 256-4532
c. ADDRESS (Include Zip Code) Commander, U.S. Army Natick RD&E Center ATTN: STRNC-IRT Natick, MA 01760-5019	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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